Distributivity and the Commensurability of Value-Definite Realism and Empirical Adequacy

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Abstract

When does an empirically adequate theory admit a value-definite completion — an assignment of simultaneous definite values to all observables? The answer depends on the algebraic structure of the observations. For a directed system of Boolean algebras carrying compatible charges, the Stone construction produces an unconditional measure on the dual space, and every observable admits a simultaneous definite value via ultrafilters. The passage to a probability measure on an underlying state space requires only σ-additivity, and the choice of state space is unconstrained by the algebra [9].

For orthomodular lattices — the algebraic framework for incompatible observations — the situation is structurally different. The McDonald–Bimbó dual space carries principal filters as distinguished structure, and the Kochen–Specker theorem blocks global value-definiteness. Probabilistic realism (descent to a probability measure) remains available via Gleason-type results, but value-definite realism does not.

The dividing line is distributivity. The state space admits a natural filtration $\mathcal{S} \supseteq \mathcal{S}_\sigma \supseteq \mathcal{S}_{\mathrm{df}}$, whose three transitions — pasting, regularity, sharpness — are of different mathematical character. Distributivity is the exact condition under which the filtration degenerates: all three obstructions vanish, and value-definite realism becomes commensurable with empirical adequacy.

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1 Introduction

A central question in the philosophy of science asks when an empirically successful theory can be completed to a realist one — one that assigns definite properties to an underlying reality, not merely adequate predictions to observable phenomena. Van Fraassen [11] argues that science need only aim at *empirical adequacy*: agreement with all observable phenomena. The scientific realist demands more — an underlying state space whose properties generate the observations.

This debate is usually conducted in philosophical terms. We give it a mathematical formulation.

Three nested positions, each a strengthening of the previous:

(i) **Empirical adequacy (EA).** A measure on the dual space of the observation algebra that agrees with all observed charges. No commitment to an underlying state space.

(ii) **Probabilistic realism (PR).** A probability measure on an underlying state space $(\Omega, \sigma(\mathcal{C}))$ that generates the observations. Requires descent from the dual-space measure.

(iii) **Value-definite realism (VDR).** Every observable simultaneously has a definite value. A global dispersion-free state — equivalently, a 2-valued homomorphism on the full observation algebra.

These are nested: VDR implies PR implies EA. Which implications reverse depends on the observation algebra:

Compatible observations. When all observations can be performed simultaneously, the observation algebra is a Boolean algebra — a complemented distributive lattice. Every Boolean algebra admits 2-valued homomorphisms (ultrafilters exist by Zorn's lemma). In this setting, all three positions are available and mutually commensurable. The passage from EA to PR requires σ-additivity of the charges [9]; the further passage to VDR is free.

Incompatible observations. When some observations cannot be performed simultaneously — as in quantum mechanics, contextual experimental design, or distributed systems with interfering sensors — the observation algebra is an orthomodular lattice (OML), a complemented but non-distributive lattice. The Kochen–Specker theorem [8] shows that for the prototypical case $L(H)$ with $\dim H \geq 3$, no 2-valued homomorphism exists: VDR is blocked. Probabilistic realism remains available via Gleason's theorem [5], and empirical adequacy is unconditional.

The dividing line is distributivity. We begin with the Boolean case [9].

Let $(\iota, \leq)$ be a directed set indexing a family of Boolean algebras $\{B_i\}_{i\in\iota}$ with injective homomorphisms $\varphi_{ij} : B_i \to B_j$ for $i \leq j$. Write $\mathcal{C} = \bigcup_i B_i$ for the direct limit — the *cylinder algebra.*

A *compatible family of normalised charges* is a collection $\{\ell_i\}$ with $\ell_i : B_i \to [0, 1]$ finitely additive, $\ell_i(1) = 1$, and $\ell_j \circ \varphi_{ij} = \ell_i$ for $i \leq j$.

The Stone space $\mathrm{St}(\mathcal{C})$ is the space of ultrafilters on $\mathcal{C}$, compact and totally disconnected. Each $\omega \in \Omega$ determines a principal ultrafilter $\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$.

Any compatible family of normalised charges extends to a σ-additive Baire probability measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ — unconditionally, with no assumption of σ-additivity [3, Theorem 1], [9]. This is the *empirically adequate* object: it agrees with all charges on cylinder events.

The measure $\hat{\mu}$ lives on $\mathrm{St}(\mathcal{C})$, not on Ω. Descent to a probability measure P on $(\Omega, \sigma(\mathcal{C}))$ requires $\hat{\mu}(\mathrm{pure}(\Omega)) = 1$ — that $\hat{\mu}$ concentrates on the principal ultrafilters. This holds if and only if each charge ℓ_i is σ-additive [9].

Every Boolean algebra admits ultrafilters (by Zorn's lemma), and every ultrafilter is a 2-valued homomorphism. A 2-valued homomorphism assigns to each element $a \in \mathcal{C}$ a definite value in $\{0, 1\}$, respecting the lattice operations.

Moreover, any subset $S \subseteq \mathrm{St}(\mathcal{C})$ that projects surjectively onto each $\mathrm{St}(B_i)$ can serve as a realisation space: set $\Omega = S$ with evaluation maps given by the restriction maps. The algebra imposes no constraint on the choice of Ω [9].

Proposition 1.1 (Boolean commensurability)**.** *For a directed system of Boolean algebras with compatible charges:*

(a) EA is unconditional: $\hat{\mu}$ *on* $\mathrm{St}(\mathcal{C})$ *always exists.*

(b) PR holds iff each ℓ_i *is* σ*-additive.*

(c) VDR holds unconditionally: $\mathrm{St}(\mathcal{C})$ *consists entirely of 2-valued homomorphisms.*

The only obstruction is σ*-additivity at the EA–PR transition, and it is a condition on the charges, not the algebra.*

Proof. Part (a) is the Stone construction [3, 9]. Part (b) is [9, Theorem 1]. Part (c): ultrafilters on a Boolean algebra are exactly the 2-valued homomorphisms; their existence follows from Zorn's lemma applied to the filter generated by any non-zero element. □

2 Orthomodular observation algebras

Proposition 1.1 depends on distributivity at exactly one point: the existence of 2-valued homomorphisms. When observations are incompatible, this fails.

Definition 2.1. An *ortholattice* is a bounded lattice $A = (A, \wedge, \vee, {}^{\perp}, 0, 1)$ with an orthocomplementation ${}^{\perp} : A \to A$ satisfying $a \wedge a^{\perp} = 0, \quad a \vee a^{\perp} = 1, \quad a^{\perp\perp} = a$, and $\quad a \le b \Rightarrow b^{\perp} \le a^{\perp}$.

Definition 2.2. An *orthomodular lattice* (OML) is an ortholattice satisfying the *orthomodular law*:

$$a \le b \quad \Longrightarrow \quad b = a \vee (a^{\perp} \wedge b).$$

Every Boolean algebra is an OML (the orthomodular law follows from distributivity), but the converse fails. An OML is Boolean if and only if it is distributive [10, Proposition 2.3].

Example 2.3. The lattice $L(H)$ of closed linear subspaces of a separable Hilbert space H with $\dim H \ge 3$, ordered by inclusion with $U^{\perp} = \{x \in H : \langle x, y\rangle = 0 \ \forall y \in U\}$, is an orthomodular lattice that is not distributive [7, Chapter 1].

Example 2.4 (Incompatible measurements)**.** Three binary measurements A, B, C, where (A, B) and (A, C) can be performed jointly but (B, C) cannot. The jointly measurable pairs generate Boolean subalgebras; the full observation algebra is orthomodular, not Boolean.

Two elements a, b in an OML *commute* if they generate a Boolean subalgebra. A maximal set of pairwise commuting elements forms a *Boolean context* — a maximal Boolean subalgebra of A.

Definition 2.5. A *state* on an OML A is a function $s : A \to [0, 1]$ with $s(1) = 1$ and

$$a \perp b \quad \Longrightarrow \quad s(a \vee b) = s(a) + s(b),$$

where $a \perp b$ means $a \le b^{\perp}$.

A state is additive on *orthogonal* pairs only — not on arbitrary joins. This is the OML analogue of a finitely additive charge on a Boolean algebra.

Definition 2.6. A state s is *dispersion-free* if $s(a) \in \{0, 1\}$ for all $a \in A$.

A dispersion-free state is a 2-valued homomorphism: it assigns a definite truth value to every proposition in A. On a Boolean algebra, dispersion-free states are exactly the ultrafilters.

Theorem 2.7 (Kochen–Specker [8])**.** *If* $\dim H \ge 3$*, there is no dispersion-free state on* $L(H)$*.*

For $\dim H = 2$, dispersion-free states exist; the obstruction is specific to $\dim H \ge 3$.

Theorem 2.8 (Gleason [5])**.** *If* $\dim H \ge 3$*, every* σ*-additive state on* $L(H)$ *is of the form* $s(P) = \mathrm{tr}(\rho P)$ *for a unique density operator* ρ *on* H*.*

The analogue of Stone duality for OMLs is the McDonald–Bimbó duality [10].

Let A be an OML. A *filter* on A is a non-empty subset $x \subseteq A$ that is upward closed ($a \in x$ and $a \le b$ implies $b \in x$) and closed under finite meets ($a, b \in x$ implies $a \wedge b \in x$). A filter is *proper* if $0 \notin x$; the unique *improper* filter is $\omega = A$.

For each $a \in A$, the *principal filter* generated by a is $\uparrow a = \{b \in A : a \le b\}$. Write $F(A)$ for the collection of all filters on A, and $P(A) \subseteq F(A)$ for the subcollection of principal filters.

Definition 2.9 (McDonald–Bimbó [10])**.** Let A be an OML. The *dual space* of A is the ordered relational topological space

$$S_0(A) = \big(F(A),\ \subseteq,\ \perp_A,\ P(A),\ \mathcal{T}\big),$$

where:

(i) $F(A)$ is the set of all filters on A;

(ii) $\subseteq$ is set inclusion;

(iii) $x \perp_A y$ iff there exists $a \in A$ with $a \in x$ and $a^\perp \in y$;

(iv) $P(A)$ is the collection of principal filters;

(v) $\mathcal{T}$ is the topology generated by the subbasis $\{h(a) : a \in A\} \cup \{F(A) \setminus h(a) : a \in A\}$, where $h(a) = \{x \in F(A) : a \in x\}$.

Theorem 2.10 (McDonald–Bimbó [10]). *$S_0(A)$ is an orthomodular space (compact, totally disconnected). The category of OMLs and homomorphisms is dually equivalent to the category of orthomodular spaces and continuous weak p-morphisms. Moreover, $A \cong \mathrm{CO}(S_0(A))^\dagger$, the OML of clopen $\perp$-stable subsets of $S_0(A)$.*

The critical differences from Stone duality:

Feature	Boolean (Stone)	OML (McDonald–Bimbó)
Dual points	Ultrafilters	All filters
Distinguished subset	None	$P(A)$ (principal filters)
Clopens	Boolean algebra	OML
Orthogonality	Trivial	Non-trivial ($\perp_A$)
2-valued homomorphisms	Plentiful (Zorn)	May not exist (KS)

In Stone duality, the principal filters (i.e., the principal ultrafilters pure(ω) for $\omega \in \Omega$) are *invisible*: they play no distinguished role in the dual space. Which ultrafilters are principal depends on the choice of Ω [9].

In the McDonald–Bimbó duality, $P(A)$ is *part of the structure* of the dual space: the orthomodular frame conditions (Definition 2.9) explicitly reference $P(A)$. This is the algebraic formalisation: for non-distributive algebras, the distinction between realised and unrealised observation-patterns is constrained by the algebra. Distributivity is the condition under which this constraint vanishes.

$S_0(A)$ is compact, so the topological machinery applies. But the algebraic input is weaker. A state s on A defines $\mu(h(a)) = s(a)$ on $\mathrm{CO}(S_0(A))^\dagger$.

In the Boolean case, $\mathrm{Clop}(\mathrm{St}(\mathcal{C}))$ is a Boolean algebra of sets, and s is finitely additive on it. The Carathéodory–Choksi extension machinery applies: compactness of $\mathrm{St}(\mathcal{C})$ gives σ-subadditivity, and the extension to a Baire measure is automatic.

In the OML case, $\mathrm{CO}(S_0(A))^\dagger$ is an OML of sets — closed under $\cap$ and $^\perp$ but *not* under arbitrary $\cup$. The state s is additive on orthogonal pairs only. Carathéodory does not apply: there is no Boolean algebra of sets on which s is finitely additive.

For the prototypical case $A = L(H)$ with $\dim H \geq 3$, Gleason's theorem (Theorem 2.8) provides the extension: every state arises from a density operator, giving a σ-additive measure via the Born rule. For general OMLs, the extension problem is open.

Even when probabilistic descent succeeds, value-definite descent does not. The Bub–Clifton theorem [2] addresses what *can* be salvaged:

Theorem 2.11 (Bub–Clifton [2]). *Given a state ρ on $L(H)$ and a preferred observable R, there is a unique maximal sublattice $D(\rho, R) \subseteq L(H)$ such that:*

(i) $D(\rho, R)$ is a Boolean algebra;

(ii) $D(\rho, R)$ contains the spectral projections of R;

(iii) The restriction of ρ to $D(\rho, R)$ is a mixture of dispersion-free states.

$D(\rho, R)$ is the maximal Boolean subalgebra on which VDR is available. It depends on R: different preferred observables yield different maximal Boolean subalgebras.

3 Distributivity as the commensurability condition

The status of the three positions depends on the algebra:

$$\underbrace{\text{EA}}_{\text{always}} \xleftarrow{\sigma\text{-add.}} \underbrace{\text{PR}}_{\text{Gleason}} \xleftarrow{\text{VDR}} \underbrace{\text{dispersion-free}}_{\text{Zorn (Boolean) / KS-blocked (OML)}}$$

The rightmost arrow is free in the Boolean case and obstructed in the OML case.

Theorem 3.1 (Commensurability)**.** *Let A be a complemented lattice.*

(a) If A is distributive (Boolean), then EA, PR, and VDR are mutually commensurable:

- *EA is unconditional (Stone construction).*
- *PR holds iff the charges are σ-additive* [9]*.*
- *VDR holds unconditionally (ultrafilters exist).*

Moreover, the choice of realisation space Ω is unconstrained by the algebra.

(b) If A is non-distributive (OML with $A \cong L(H)$, $\dim H \geq 3$), then:

- *EA is unconditional.*
- *PR holds (Gleason, Theorem* 2.8*).*
- *VDR fails globally (Kochen–Specker, Theorem* 2.7*).*
- *VDR holds locally within each Boolean context (Bub–Clifton, Theorem* 2.11*).*

Proof. Part (a) is Proposition 1.1. For part (b): EA follows from the existence of states on $L(H)$ and the McDonald–Bimbó dual [10]; PR from Gleason's theorem; VDR-failure from Kochen–Specker; local VDR from Bub–Clifton. □

Corollary 3.2. *Distributivity of the observation algebra is the exact condition under which value-definite realism and empirical adequacy are commensurable.*

Proof. If A is distributive, VDR holds (Theorem 3.1(a)). If A is non-distributive with $A \cong L(H)$ and $\dim H \geq 3$, VDR fails (Theorem 3.1(b)). Since an OML is Boolean iff it is distributive [10], the dividing line is exactly distributivity. □

Remark 3.3. The corollary requires the dimensional condition $\dim H \geq 3$ for the Kochen–Specker direction. For $\dim H = 2$, the OML $L(\mathbb{C}^2)$ is non-distributive but admits dispersion-free states: VDR holds despite non-distributivity. The sharp characterisation is: VDR is commensurable with EA iff A is distributive *or* every non-distributive component has dimension at most 2. In physically relevant cases ($\dim H \geq 3$), distributivity is the effective dividing line.

The underlying structure is a filtration of the state space, whose three transitions require different keys.

Definition 3.4 (Coherence filtration)**.** Let A be a complemented lattice with a family of charges. The *state space* of A admits a descending filtration:

$$\mathcal{S}(A) \;\supseteq\; \mathcal{S}_\sigma(A) \;\supseteq\; \mathcal{S}_{\mathrm{df}}(A),$$

where:

- $\mathcal{S}(A)$ is the set of states on A (contextual coherence);
- $\mathcal{S}_\sigma(A) \subseteq \mathcal{S}(A)$ is the subset of σ-additive states (probabilistic coherence);

- $\mathcal{S}_{\mathrm{df}}(A) \subseteq \mathcal{S}_\sigma(A)$ is the subset of dispersion-free states (value-definite coherence).

Beneath the filtration sits a local condition: *consistency* requires only that each maximal Boolean subalgebra of A individually admits a compatible state.

Each inclusion can be strict. The three transitions have different mathematical character:

Transition	Character	Key	Obstruction
Consistency $\to \mathcal{S}(A)$	Pasting (topological)	Do local sections glue?	Contextuality
$\mathcal{S}(A) \to \mathcal{S}_\sigma(A)$	Regularity (analytic)	Does the state have good limits?	σ-additivity
$\mathcal{S}_\sigma(A) \to \mathcal{S}_{\mathrm{df}}(A)$	Sharpness (algebraic)	Can probabilities collapse to $\{0,1\}$?	Kochen–Specker

Distributivity collapses the filtration. For a Boolean algebra:

- *Pasting* is trivial: every local assignment extends to a global state (distributivity guarantees compatibility).
- *Sharpness* is free: ultrafilters exist (Zorn's lemma), so $\mathcal{S}_{\mathrm{df}}(A) \neq \varnothing$ unconditionally.
- The only non-trivial transition is *regularity*: σ-additivity of the charges. This is the content of [9].

For an OML with $\dim \geq 3$, every transition is non-trivial:

- *Pasting* is non-trivial: contextuality obstructions are possible.
- *Regularity* requires Gleason-type results.
- *Sharpness* is blocked: $\mathcal{S}_{\mathrm{df}}(A) = \varnothing$ (Kochen–Specker).

The filtration is thus a diagnostic: distributivity is the condition that makes it degenerate.

Remark 3.5 (Relation to contextuality hierarchies)**.** Abramsky and Brandenburger [1] introduced a hierarchy of contextuality — probabilistic, possibilistic, and strong — grading the *strength of the obstruction* to a global section of a presheaf of probability distributions. Our filtration is complementary: it grades *what the algebra can support*, not how badly a global section fails. Their axis measures obstruction strength; ours measures coherence demands.

4 Discussion

The realism debate is not uniform

The standard philosophical debate between scientific realism and constructive empiricism treats the question as uniform: either theories aim at truth or they aim at empirical adequacy. The algebraic analysis above shows that the question is *parametrised* by the observation algebra.

For Boolean observations, the debate is genuine: the mathematics does not prefer one position over another. EA, PR, and VDR are all available; σ-additivity is the only bridge condition, and it is a property of the charges, not of the algebra.

For non-Boolean observations, the mathematics is not neutral. VDR is algebraically impossible (for $\dim \geq 3$); PR is available but structurally constrained; EA is always available. The constructive empiricist position [11] is the generic one. Value-definite realism is the special case — the case in which the observation algebra happens to be distributive.

Historical echoes

The distinction among the three positions illuminates classical debates:

- *Einstein's realism* ("God does not play dice") is the demand for VDR. The Kochen–Specker theorem shows this is not merely philosophically contestable but algebraically impossible for $\dim \geq 3$.
- *Bohr's complementarity* is the observation that different Boolean contexts cannot be simultaneously realised. In our hierarchy, this is the non-triviality of the $1 \to 2$ transition.
- *Bell's beables* [6] — the observables that *can* have definite values — are exactly the elements of the maximal Boolean subalgebra $D(\rho, R)$ identified by Bub–Clifton (Theorem 2.11).
- *De Finetti's operationalism* is EA in the Boolean case: probability as coherent betting rates on observables, without commitment to an underlying state space.

Beyond quantum mechanics

The analysis applies wherever observations are incompatible — not only in quantum mechanics. Orthomodular structure arises in contextual behavioural science, distributed systems with interfering sensors, database queries with access restrictions, and experimental designs with mutually exclusive protocols. In each setting, the question "does empirical adequacy entail value-definiteness?" has the same algebraic answer: only if the observation algebra is distributive.

Open questions

Several directions remain:

1. *The OML extension problem.* Does every state on a general OML extend to a σ-additive measure on $S_0(A)$, as in the Boolean case? Gleason's theorem answers this for $L(H)$; the general case is open (§2).
2. *Intermediate algebras.* Between Boolean algebras and arbitrary OMLs lie the MO_n-free OMLs, modular ortholattices, and other subvarieties. Where exactly does VDR fail in this spectrum?
3. *Dynamical structure.* If the observation algebra carries a group action (stationarity, exchangeability), how does symmetry interact with the coherence hierarchy? The Boolean case is partially understood (de Finetti's theorem for exchangeability); the OML case is unexplored.
4. *The topos connection.* The Döring–Isham topos approach [4] addresses closely related questions via spectral presheaves rather than the McDonald–Bimbó filter-space duality. A comparison of the two frameworks — and whether the commensurability theorem has a topos-theoretic formulation — is a natural next step.

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